DO NOW — RECALL

A*: heuristic search I

Name: _____ Class: _____ Date: _____

1. Why are predecessors recorded when a distance improves?

2. Can Dijkstra assume negative edge weights are safe?

3. What is 2+3+2?

4. Starting S=0, edge S–A=4, then A–G=2. What is the S→A→G path cost?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

They allow reconstruction of the actual shortest route.

2. Model answer

No. Its shortest-path guarantee requires non-negative weights.

3. Model answer

7.

4. Model answer

6.

Vocabulary — know these terms

g(n) — Known best cost from the start to vertex n.

h(n) — Estimated cost from vertex n to the goal.

f(n) — Combined score g(n)+h(n).

frontier / open set — Discovered vertices awaiting expansion.

admissible heuristic — An estimate that never exceeds the true cheapest remaining cost.

NEW LEARNING

Read, annotate and explain

g, h and f have different meanings

g is distance already travelled; h estimates the remaining distance; f=g+h is the estimated total.

Start with a frontier

From S, A gets g2,h3,f5, while B gets g6,h1,f7. Choose A, not simply the smallest h.

Improve a discovered route

From A, B improves to g4,h1,f5. From B, G improves to g6,h0,f6.

Stop on the selected goal

When G is selected from the frontier, reconstruct S→A→B→G with total cost 6.

Keep the heuristic safe

The estimates h(S)=5,h(A)=3,h(B)=1,h(G)=0 never exceed the actual cheapest remaining cost.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace A* on P, Q, R, G using the supplied admissible estimates.

Predict the next state

From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.

Update the table

Through Q, R improves to g4,h2,f6; G gets g11.

Update the table

From R, G improves to g6,h0,f6.

Update the table

Select G: predecessor chain yields P→Q→R→G, cost 6.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A node has g=7 and h=3. Give f and define each value.

Question 2

Graph edges S–A=2, S–B=5, A–B=1 and B–G=2. Heuristics: h(S)=4,h(A)=2,h(B)=1,h(G)=0. Calculate f for A and B when first found and state the optimal S→G route.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why admissibility matters. For the graph S–A=2, A–G=4 and S–G=10, what is the greatest admissible h(A), and why?

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A* chooses the next vertex solely by its smallest heuristic h(n).

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.